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The fruit fly is a species of small insect.

The fruit fly has a gene that codes for an enzyme called alcohol dehydrogenase (AD). AD catalyses the breakdown of alcohol when alcohol is in the insects' food.

The gene coding for AD has two alleles, **AD^F** and **AD^S**.

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The enzyme encoded by the **AD^F** allele catalyses the breakdown of alcohol **faster** than the enzyme encoded by the **AD^S** allele. Suggest why.

[3 marks]

A scientist took a random sample of adult fruit flies from a population. He measured the frequency of the **AD^F** allele in this sample (generation 0). He then:

- selected 100 of these insects at random and kept them in a container
- fed the insects food containing alcohol
- let the insects reproduce
- repeated these steps for 45 generations of fruit fly reproduction.

The scientist measured the frequency of the **AD^F** allele in the 45th generation.

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Suggest why the scientist took his sample from the population at random.

[1 mark]



Table 2

Generation of fruit fly reproduction	Frequency of AD ^F
0	0.20
45	0.74

Alcohol is toxic to fruit flies. Suggest and explain why the frequency of the **AD^F** allele changed during the 45 generations.

[4 marks]

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Identify the type of selection investigated in the 45 generations of fruit fly reproduction. Tick (✓) **one** box.

[1 mark]

- No selection

11

- ### Directional selection

1

- ## Random selection

1

- ## Stabilising selection

11

